Maaz Qureshi

Robotics Software Engineer (Physical AI and ML) — MASc Mechatronics, University of Waterloo

Portfolio | LinkedIn | GitHub | Google Scholar | +1 437 870 8791 | m23qures@uwaterloo.ca | Waterloo, Canada

Skills

Languages & Frameworks: Python | C++ (Multi-threading, OOP) | PyTorch | TensorFlow | TensorRT (Nvidia CUDA) | Eigen/Numpy
Machine Learning: CNN | VLA | LLM | Quantization | Training (Backpropagation) | GPU Inference | Reinforcement Learning (applications)
Robotics: VI SLAM (Volumetric Mapping & Localization) | Perception Vision Models (deep learning) | ADAS (Driving) | Sensor Fusion (3D Stereo RGB-D Camera, LiDAR, mmWave 4D RADAR, IMU) | Computer Vision (OpenCV) | Motion Planning 7-DoF (IK/FK) | ROS2 | 5G
Tools: NVIDIA CUDA | Docker CI/CD | | gRPC | RViz | GTSAM | Gazebo | ORBSLAM3 | Software QA (Debugging: Valgrind, GDB; Monitoring: Jtop)

Experience

Software Engineer (Physical AI): Project Lead — April. 2026 – Present
Playroom Inc — *Bay Area California, USA and remote Canada*
- Project lead for Machine Learning (VLAs/Imitation learning) for Robotics (physical AI) and World models from data generated from (i) Human Teleoperation and (ii) Game Environment.

Research Engineering (research RA with University of Waterloo) — Present
Apple — *Waterloo, Canada*
- Confidential-NDA project. Leading the research on VLA learning Software Pipeline for Apple Robotics project.

Robotics Software Engineer: — Sept. 2025 – April 2026
Cobionix Corporation Inc — *Kitchener, Canada*
- Built Perception deep learning AI models (pose estimation, object tracking, and detection) C++ nvidia CUDA GPU for real-time RGB-D.
- Developed live RGB-D stereo sensor fusion algorithm with intrinsic/extrinsic (temporal and spatial 90%+) calibration pipeline.[Link]
- Optimization for TensorRT models for async pipelines to maximize throughput (increase camera FPS, reduce latency) to run parallel AI.

Graduate Research Student: Robotics Perception Software Development — Jan. 2024 – Sept. 2025
University of Waterloo RoboHub Lab [Preprint Publication, Video, GitHub] — *Waterloo, Canada*
- Developed distributed perception (vision) pipeline with mmWave 4D Automotive RADAR + RGB-D, achieving 92% accuracy.
- Developed novel multi-robot dense volumetric SLAM for SMART factory robotics (C++/ROS2 humble) with TSDF-octree + ORBSLAM3, achieving 85% improvement over state-of-the-art baselines for high-fidelity 3D scene reconstruction with 4 AMRs (mobile robots).

Keysight Technologies Inc [Preprint Publication, Video, GitHub]
- Developed motion planning (OMPL + RRT) for 7-DoF Franka/KUKA collaborative robots SIM-to-Real deployment for SMART factory.
- Achieved 100% joint-limit and collision avoidance across hemispherical pattern formations. Used python with ROS rviz, and docker setup.

Rogers Communications Inc | MITACS Research Thesis Internship
- Integrated 5G in DISTRIBUTED software pipeline on NVIDIA Jetson for smart factory robots achieved 37% latency reduction.
- Benchmarked 5G sub-6 GHz (3.5 GHz) and mmWave (28 GHz) channels against WiFi for industrial robotics to increase bandwidth.

Asset Integrity Engineer: Robotics Software — May 2022 – Dec. 2023
ABYSS Solutions Inc — *Islamabad (Onsite) & Australia*
- Worked Visual-SLAM (MAPPING) models for mooring chain autonomous robot mapping and 3D digital twin development.
- Contributed to software development for ROS/ROS2 AMR pipelines, VR for ADIPEC, and agriculture quadcopter projects. [Link]
- Led a 10-engineer team on digital inspection, 18,000+ assets with 91% coverage for a BP offshore rig and Dam Data-Capture.[Link]

Side Projects

- **Humanoid Robotics – Non-Verbal Human-Robot Interaction** [Link]: NAO robot in kitchen with humans for breakfast preparations.
- **Visual Features with Machine Learning:** KNN, SVM, and FRIQUE image quality benchmarking for computer-generated vs real-world data.
- **AMRs in Health Care** [Link]: RFID nursing robot for pill dispensing, perception, and live stream in healthcare environments.

Education

University of Waterloo | Thesis Research: AMRs 3D SLAM Link1 , 7 DoF Manipulator Motion Planning Link2 — Waterloo, Canada
Master of Applied Science (MASc), Mechatronics Engineering ***(CGPA: 93.5%)*** — January 2024 – September 2025
Supervisor(s): Dr. William Melek ; Dr. George Shaker

Bahria University | Thesis: AMRs Swarm Robotics Autonomy and Perception — Islamabad, Pakistan
Bachelor of Science (BSc), Electrical Engineering ***(CGPA: 80.5%)*** — September 2017 – August 2021

Selected Publications

- **M. Qureshi et al.**, "CRADMap: Applied Distributed Volumetric Mapping with 5G-Connected Multi-Robots and 4D Radar Perception," *IEEE ICARM 2025*. [Link]
- **M. Qureshi et al.**, "Automated Angular Received-Power Characterization of Embedded mmWave Transmitters Using Geometry-Calibrated Spatial Sampling," *IEEE Transactions on Microwave Theory and Techniques (T-MTT)*, 2026. [arXiv]

Achievements

- **Employee Recognition:** Best Performer Employee of the Month, Abyss Solutions Inc (industry).
- **Innovation Awards:** Winner of 3 hackathons in robotics and VR innovation at Abyss Solutions.
- **Academic Excellence:** Awarded full MASc graduate funding and recognized among the top 3 undergraduate thesis projects.